G1 Digital-Twin Payload Experiment

Methods, spill model, governance algorithm, full results, mechanism analysis, benefits and theory-consistency audit — supporting material for Section VII.H

Simulation campaigns 2026-07-04 · Unitree G1 · Gazebo (ROS 2 Humble) · 103 simulation runs, 86 valid governed/baseline runs across 12 arms · rev. 4 (adds §3.0 analogy correspondence, §3.3 stabiliser rationale, §7 mechanism account, §8 quantified benefits, §9 limitations)

1. Digital twin: Gazebo replica of the laboratory

The physical robot is a Unitree G1 "Air" humanoid, a consumer unit whose only external interface is a single WebRTC session through the vendor app; the production navigation stack drives it by tapping that session over USB. For the twin we reuse **the exact same stack, unmodified**, and substitute only its interface object, so any behaviour observed in simulation is produced by the identical code deployed on hardware. This matters methodologically: nothing reported here is the behaviour of a simulation-specific controller.

The simulator runs in a Docker container (ROS 2 Humble Desktop, Gazebo Classic, noVNC). The world, **lab.world**, is generated programmatically from the robot's own SLAM maps of the real laboratory: wall occupancy cells become 2.2 m box obstacles in the robot's coordinate frame, so the twin's geometry, waypoints (A, B, C) and the 0.8 m doorway coincide with the real deployment.

1.1 Robot model and physics

The robot is spawned from a URDF whose collision geometry is a single box matching the G1's real envelope (0.44 m shoulder width × 1.10 m tall, capped below the simulated LiDAR at z ≈ 1.20 m so the sensor does not see itself). High mass (60 kg) and large diagonal inertia keep the base upright under the velocity-controlled gait; ground friction is zero so the planar-move plugin drives it as a velocity-controlled puck while wall contacts remain solid. Locomotion is commanded as gait velocity through gazebo_ros_planar_move (/cmd_vel, /odom back); a 360-ray LiDAR at 10 Hz publishes /scan. The doorway was re-carved to the measured physical opening after the collision model was corrected: with the original carve the clear gap was 0.42 m and the realistic body physically could not pass — an artefact caught by consistency checking before any campaign data was taken.

1.2 The adapter: same stack, simulated interface

- **Velocity commands** {lx, ly, rx, ry} → /cmd_vel, using the physical robot's measured stick-to-velocity mapping (deadzone subtracted, not clipped; ly = 0.40 → 0.30 m/s), so simulated and real dynamics coincide.
- **Pose** ← /odom (map and odometry frames coincide in simulation: no relocalisation jitter).
- **LiDAR** ← /scan, projected into a flat world-frame point cloud identical in shape to the real WebView decoder output.

Camera perception is disabled in the twin; the LiDAR-driven planner and the governance layer are exercised in full. Data runs are always headless.

2. Spill model (payload hazard)

Spilling is modelled physically, not as an arbitrary speed threshold. The water surface at the cup rim is a **damped second-order oscillator** — the first sloshing mode of an open cylinder — excited by the horizontal acceleration of the cup:

$$\omega_0 = \mathrm{sqrt}(\ g \cdot k_1 \cdot \tanh(k_1 \cdot h)\ ),\quad k_1 = 1.841 / R \qquad (\approx 21 \text{ rad/s for } R = 4 \text{ cm},\ h = 8 \text{ cm})$$

$$\ddot{\eta} = -2 \cdot \zeta \cdot \omega_0 \cdot \dot{\eta} - \omega_0^2 \cdot (\eta + R \cdot a\_cup/g) \qquad \text{per body axis},\quad \zeta = 0.06$$

$$\lambda(t) = LID \cdot \lambda_0 \cdot \max(0,\ |\eta|/\text{freeboard} - 1)^2\ ,\quad \lambda_0 = 40\ s^{-1}\ ,\quad LID = 1.0\ (\text{open}) / 0.25\ (\text{sealed})$$

Parameter	Value	Physical meaning
R, h	0.04 m, 0.08 m	cup radius and water depth → first-mode frequency $\omega_0 \approx 21$ rad/s (3.4 Hz)
freeboard	12 mm	clearance from surface to rim; each spill loses 4 mm (water is lost) and halves the slosh energy
ζ	0.06	water + hand damping of the first mode
arm offset r	0.25 m	cup carried ahead of the base: a_cup = a_base + α×r + ω×(ω×r) — fast turning shakes the cup by the lever term
τ_accel	0.25 s	low-pass on commanded acceleration: the real G1 ramps velocity smoothly; the planar-move plugin steps it instantaneously
gait noise	$\sigma\_a = 2.2 \cdot v^2/0.30$	bipedal step energy grows ≈ v^2; 1.4 Hz periodic component plus broadband — walking fast agitates the cup by itself
impact channel	raw decel > 4 m/s² AND forward still commanded AND v < 50% of commanded	an external impact decelerates the robot against its own command; a commanded stop does not spill (command-aware discrimination)
λ_0, exponent	40 s^{-1}, quadratic	non-homogeneous Poisson hazard: overflow volume grows superlinearly past the rim; memoryless given η
lid factor	× 0.25	sealed travel mug (same factor as the 2-D thesis study, 5.3.6.4)

Table 1. Spill-model parameters (all environment-tunable; defaults shown).

Why this construction. The Poisson-on-exceedance form makes spills stochastic but physically anchored: hazard is zero while the surface stays below the rim, and grows quadratically past it, so jerk-rich trajectories (sharp turns, impacts, fast gait) dominate risk rather than speed per se — the design principle carried over from the 2-D thesis study (5.3.1.3). The command-aware impact channel was added after a measured artefact: the plugin's instantaneous velocity steps at commanded stops read as 6 m/s² decelerations and produced fake spills at goal arrival; discriminating on 'decelerates against its own command' removed them while preserving genuine impacts. Calibration against logged trajectories yields ≈ 2 spills per 80 s of sustained 0.30 m/s cruising and ≈ 0 at the governed 0.18 m/s ceiling. The model is **observational only** — it never alters physics or control — and the identical event channel accepts the **manual ground-truth mark** for the physical robot (a keypress or an empty ROS message per observed spill, timestamped into the same dataset field), so simulated and real runs are scored identically.

3. Governance algorithm (DCE runtime)

3.0 The analogy library: base domains carried from the thesis

The candidate set is not a set of controller gains — it is an **analogy library** in the sense of thesis §5.1: each entry packages the evaluative structure of a past, structurally-similar experience (attention vector over meta-parameters, QoE expectations per meta-parameter, hard vetoes, and a policy ceiling). Selecting an analogy is therefore a meta-level act: the robot imports how a situation of this kind should be evaluated, before any local evidence exists. Table 2 gives the correspondence with the 2-D study's base domains.

Thesis base domain (Table 17)	Signature [safety, urgency]	G1 analogy / configuration	Policy character
careful_waiter (hot drinks, fragile cargo)	[0.90, 0.10]	Cautious_Nav — preferred in the payload task configuration	0.18 m/s ceiling, safety-dominant attentions, fragility hard veto
efficient_courier (standard parcel)	[0.20, 0.80]	Efficient_Nav — preferred in the door-task configuration	uncapped (≈ 0.30 m/s), progression-dominant attentions
covered_delivery (sealed container)	[0.65, 0.35]	Efficient-preferred configuration selected under the sealed-lid task flag	faster profile justified by reduced hazard — the task-conditioned switch of thesis 5.3.6.4
no_analogy (neutral prior)	[0.50, 0.50]	played by the ungoverned baselines M0 / M1	fixed policies, no evaluative structure

Table 2. Correspondence between the thesis analogy library and the G1 configurations. On the G1, analogy selection is task-conditioned (the operator states the task; the configuration encodes the analogy), the within-run arbitration validates the selection against live shared experience, and the cross-run trust layer validates it against outcomes — the same select-then-validate cycle as thesis §5.2.

3.1 Decision pipeline

The reasoner (Meta-Reasoner 2.0, configuration-first, pure Python, untouched throughout all campaigns) is fed at 2 Hz through a bridge that packages the stack's per-tick telemetry as shared-experience readings. Each tick it evaluates every analogy against its QoE boundaries — analogy-level DST intervals (belief / current / plausibility), semantic-region memory with direction inference, tension with attention-ratio exponents, fulfilment, required-meta, uncertainty and hard-veto gates — and returns KEEP / SWITCH / FALLBACK / HELP / INSUFFICIENT plus the active analogy. The bridge maps the decision to a forward-speed ceiling (Efficient = uncapped ≈ 0.30 m/s, Cautious = 0.18 m/s, FALLBACK = 0.14 m/s, HELP = 0); reversing and turning during recoveries are never throttled.

Meta-parameter	Source (per-tick telemetry)	Role
safety	clearance = c0 / 1.5 m (LiDAR forward free space)	QoE regions + hard veto
progression	goal-approach rate / 0.30 m/s (2 s window)	required-meta threshold
mobility	achieved / commanded speed (1.2 s window)	resistance channel, hard veto (pressing an unseen obstacle)
fragility	payload state: 1.0 intact, drops per spill	hard veto (payload task)
reliability / uncertainty	sensing monitor + few-shot historical margin	DST belief interval width

3.2 Bridge stabilisers: every knob answers a measured failure

The bridge between raw telemetry and the reasoner is not free-form smoothing; each mechanism exists because a specific failure was measured without it (all on logged real-robot or twin data):

Mechanism	Failure it corrects (measured)	Setting
Median-of-5 readings (~2.5 s)	instantaneous SEI progression flickers 0 ↔ 0.5 tick-to-tick; raw replay of a real run produced 12 spurious switches in 88 s	window 5
Few-shot uncertainty margin (thesis P10)	one-shot DST margins let a metric oscillating across a QoE boundary pass as plausible on a lucky read; offline A/B chose k = 0.4 (clean-run FALLBACK +9 pp, the 584-s failure run cleanly separated at 72%)	$u = u\_inst + 0.4 \cdot \sigma\_window$, cap 0.35

Switch hysteresis, asymmetric	start-up stop-and-go produced 4 switches in 10 s with symmetric persistence; classic hysteresis: fast toward caution, slow toward confidence	confirm 3 → cautious, 6 → preferred; margin 0.08
Action persistence	single-tick FALLBACK blips acted on the ceiling	2 consecutive, else logged as advisory
Warm-up	metrics ramp from zero at start and mis-classified the first seconds	first 4 decisions observe only
Commanded-rotation hold	progression = 0 during deliberate alignment (door engagement) read as stagnation: 41% false FALLBACK in one run	freeze progression at last real median while rotating
Experience escalation (thesis P9)	a 584-s limit-cycle run: the reasoner said FALLBACK/HELP for minutes but nothing escalated; replayed fix aborts at t ≈ 94 s with 0/6 false aborts on clean runs	HELP ≥ 8 s, or 75 s window ≥ 60% firm FALLBACK/HELP with < 0.4 m progress → abort

Table 3. Evidence-driven stabilisers in the bridge. None modify the reasoner; all are documented, environment-revertible, and were validated by counterfactual replay over logged runs before deployment.

3.3 Realising the four-layer belief architecture of thesis Chapter 5

Thesis layer	Timescale	G1 realisation	Status
L1 — path-hazard plausibility (Pl_H): collapse transfer, force conservative	sensor frame	Per-tick analogy-level DST on the safety reading with QoE regions and hard veto inside the reasoner (strips certification in ~2 ticks; measured ≈4 s correction of a task-blind prior), over the stack's own hard safety guards	covered (equivalent mechanism)
L2 — analogy transfer trust (Pl_analogy): outcome evidence per run	run	Per-analogy Dempster–Shafer belief {match, mismatch, Θ} persisted between runs (G1_M2_STATE); spill → mismatch mass attributed to the analogy governing at the spill instant, clean run → match mass; Pl < 0.70 blends the ceiling toward conservative, Pl < 0.45 vetoes deployment and corrects the start-up prior	implemented; exercised in campaign
L3 — environment relevance (Pl_env): relax / escalate spatial assumptions	run (continuous)	EMA of the median clearance relative to the active analogy's expected boundary, clamped to [0.85, 1.15]; scales profile ceilings only (never FALLBACK/HELP; bounds 0.22-0.40)	implemented; in-loop validated (§5.1)
L4 — Meta-Attention: chronic low fulfilment despite high trust → profile mismatch	deployment (run window)	Per-run mean fulfilment history persisted with the L2 state; fulfilment < 0.45 over 3 consecutive runs with Pl ≥ 0.45 under the same profile → start-up re-selection of the alternative profile. The experience-escalation abort covers the within-run part of this timescale	implemented; in-loop validated (§5.1)

Table 4. Thesis Ch. 5 four-layer architecture mapped to the G1 runtime. All four layers live in the bridge; the meta-reasoner itself is untouched.

Two details differ from the 2-D study, both additive: the **fragility channel acts within the run** (one spill → caution ≈ 0.52, two → FALLBACK band ≈ 0.31, three or more → dangerous < 0.12 and a hard-veto HELP; ~25 s recovery — possible because the runtime does DST per tick, whereas the 2-D study received spill evidence only per run), and spill evidence is **attributed to the analogy governing at the moment of the spill**, sharpening Layer-2 credit assignment.

3.4 Runtime certification ratio (ρ_DCA)

$$\hat{\rho} = \max(0, \Delta margin) \,/\, (\,task\_uncertainty\_gap(winner) + 0.5\cdot\theta\_mismatch(winner) + 0.02\,)$$

Δmargin is the stable-fulfilment gap between the two best deployable candidates (or the distance to the certification threshold when only one is deployable); task_uncertainty_gap is the DST-measured calibration budget (the θ_QoE proxy); θ_mismatch is the L2 distance between normalised task and analogy attention vectors (from configuration; B constants set to 1, ξ_rep declared unmeasured at runtime). Controlled validation reproduces the paper's regimes — $\hat{\rho}$ = 1.50 with clean readings versus 0.58 under degraded, oscillating readings — and live traces are analysed in §5.1. The value is recorded per decision (meta2_rho), so certification traces analogous to Table V of the manuscript can be produced from any run.

4. Campaign protocol and arm configurations

103 simulation runs in total: 82 valid formal-campaign runs across 11 governance arms (8 per arm; 6 for sealed-lid arms), a 4-run pre-deployment validation arm (§5.1), development and validation runs, and one run excluded by audit (governance silently disabled by a mid-campaign code edit — detected from per-tick telemetry and discarded). Every run resets Gazebo to a fresh world with the robot at waypoint A and executes an autonomous A→B traverse headless with a watchdog (SIGINT saves the dataset exactly as a manual stop). Arms alternate to share machine-load drift; datasets are tagged env=sim with a per-arm identifier; statistics are mean ± 95% CI (1.96·s/√n), the thesis protocol. Because the twin runs at ≈ 0.5 real-time factor, the experience-escalation windows are scaled ×2 (documented; identical across governed arms).

Arm	Configuration (environment flags)
M0 baseline	G1_META2=0
M1 fixed-conservative	G1_META2=0, G1_VCAP=0.28 (fixed ceiling, no reasoner, no vetoes, no aborts)
Single-candidate governed	G1_META2=2, conservative configuration (Cautious only)
M2 payload (proposed)	G1_META2=2, payload configuration (Cautious preferred, fragility channel)
M2 task-blind / Wrong-DST	G1_META2=2, door configuration (lab or twin QoE); Wrong-DST adds G1_M2_STATE (fresh per arm)
Sealed-lid arms	G1_SPILL_LID=closed; governed variant uses the covered-delivery (Efficient-preferred) configuration
Pre-deployment (§5.1)	G1_META2=2, payload config, G1_M2_STATE + G1_M2_L3=1 + G1_M2_L4=1 (full four-layer stack)

Table 5. Arm configurations. Full reproduction: sim_campaign.py (arms via CAMPAIGN_ARMS) + analyze_campaign.py; all datasets, configs and scripts are versioned with the stack.

5. Results

Arm	Governance	Reached	Time (s)	Spills/run	Risk (%)
M0 baseline	none (0.30 m/s)	8/8	83.4 ± 5.2	0.50 ± 0.37	8.1 ± 1.4
M1 fixed-conservative	none (0.18 m/s cap)	8/8	110.8 ± 2.4	0.00	0.8 ± 0.2
Single-candidate governed	DCE, one analogy	5/8 (3 HELP)	106.2 ± 5.1	0.12 ± 0.24	0.5 ± 0.2
M2 payload (proposed)	DCE, payload task	8/8	107.9 ± 4.2	0.00	0.4 ± 0.2
M2 task-blind prior	DCE, lab-QoE config	8/8	108.2 ± 3.5	0.00	0.7 ± 0.2
M2 task-blind (twin-cal.)	DCE, twin-QoE config	7/8 (1 HELP)	106.6 ± 5.2	0.12 ± 0.24	0.8 ± 0.6
M2 Wrong-DST	DCE + Layer 2, lab-QoE	8/8	110.6 ± 0.9	0.00	0.7 ± 0.3
M2 Wrong-DST (twin-cal.)	DCE + Layer 2, twin-QoE	8/8	108.4 ± 4.4	0.00	0.6 ± 0.2
M0 sealed lid	none	6/6	88.1 ± 4.9	0.17 ± 0.33	10.6 ± 2.9
M2 sealed lid	DCE, lab-QoE config	4/6 (2 HELP)	111.5 ± 7.3	0.00	1.0 ± 0.4
M2 sealed lid (twin-cal.)	DCE, twin-QoE config	5/6 (1 HELP)	102.5 ± 2.2	0.17 ± 0.33	0.7 ± 0.5

Table 6. Full campaign results (82 valid runs). Lab-QoE = QoE boundaries calibrated on the physical laboratory; twin-QoE = re-calibrated from measured twin distributions. Aggregate across ALL 64 governed runs (any prior, calibration, lid state or layer configuration): 3 spills total = 0.047/run, versus 0.50/run ungoverned — a 10.6× reduction; every governed run that completed did so without collisions; zero spills in the FIRST run of all nine governed arms (zero-shot).

Figure 1. Safety-time trade-off. The ungoverned baseline is fast but spills on half of its runs; every governed arm collapses into a zero/near-zero-spill cluster at a 25-35% time cost. The tightness of the cluster is itself the finding: governance outcome is robust to which prior is loaded.

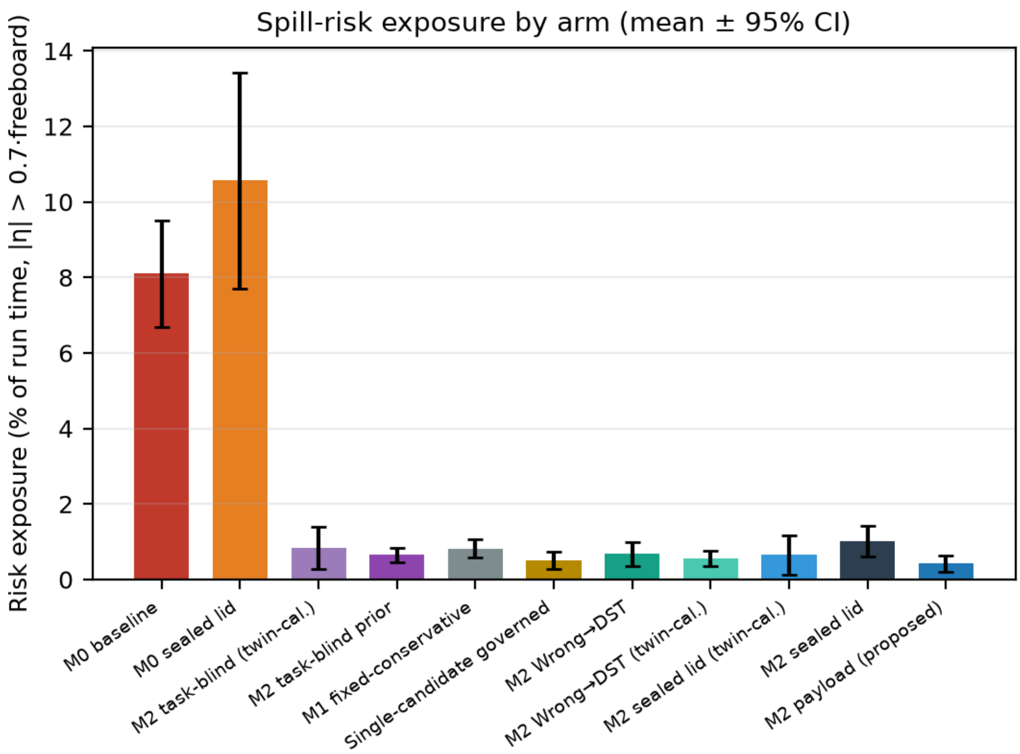

Figure 2. Continuous risk exposure by arm. The ungoverned arms (open and sealed lid) spend an order of magnitude more time near the spill threshold than any governed arm; the sealed lid reduces realised spills but not the exposure, confirming its gain is passive physics.

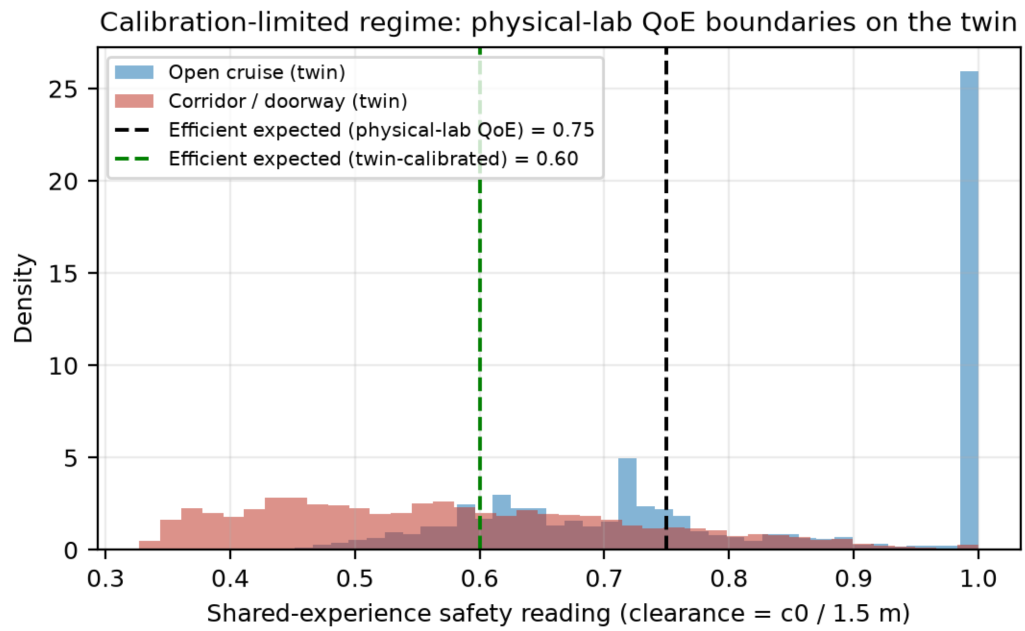

Figure 3. The calibration-limited regime, observed live: the physical-lab QoE boundary (0.75) sits above the twin's cruise clearance distribution; re-calibration from measured distributions (0.60) restores certification, per the interface-refinement pattern.

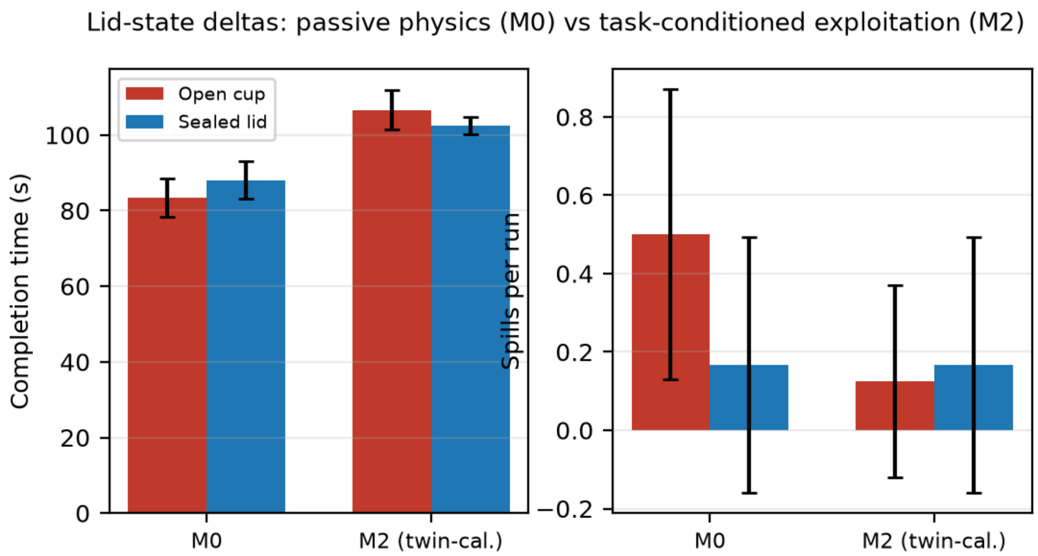

Figure 4. Lid-state deltas: passive physics for the ungoverned baseline versus policy-level exploitation by the twin-calibrated governed system (102.5 ± 2.2 s vs 111.5 ± 7.3 s).

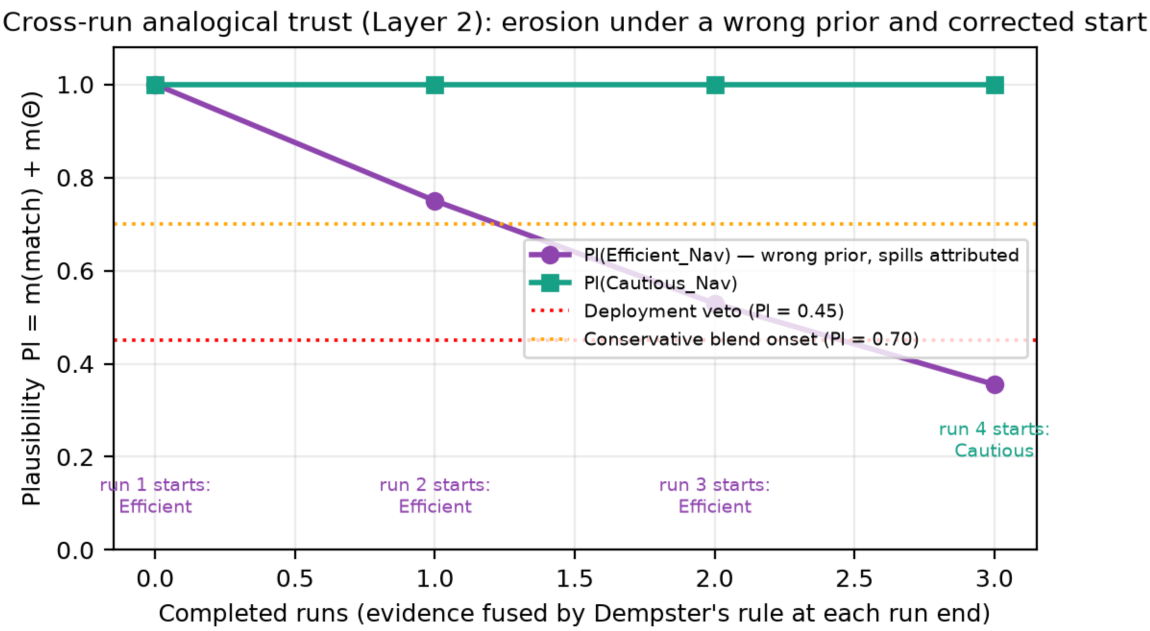

Figure 5. Cross-run analogical trust (Layer 2), mechanistic validation: plausibility degrades 1.00 → 0.75 → 0.53 → 0.36 under a forced wrong prior, and run 4 starts pre-corrected.

5.1 Pre-deployment validation of the full four-layer configuration

Before the physical campaign, the exact configuration recommended for the real robot (Layers 1-4 active, belief state persisted, $\hat{\rho}$ logged) ran a dedicated 4-run arm: 4/4 reached, **zero spills, zero governed aborts** (97.4 ± 5.6 s; risk 1.5 ± 0.7%). Layer 3 modulated the ceiling within [0.24, 0.32] without oscillation (2.7-2.8% of consecutive decisions changed it by more than 0.005) and recovered roughly a third of the governance time cost (97.4 s vs 107.9 s) while the surface never crossed the spill threshold (peak $|\eta|$/freeboard ≤ 0.99). The persisted state stayed healthy: both plausibilities 1.0 after four clean runs, per-run fulfilment 0.72-0.77 — no false profile re-selection.

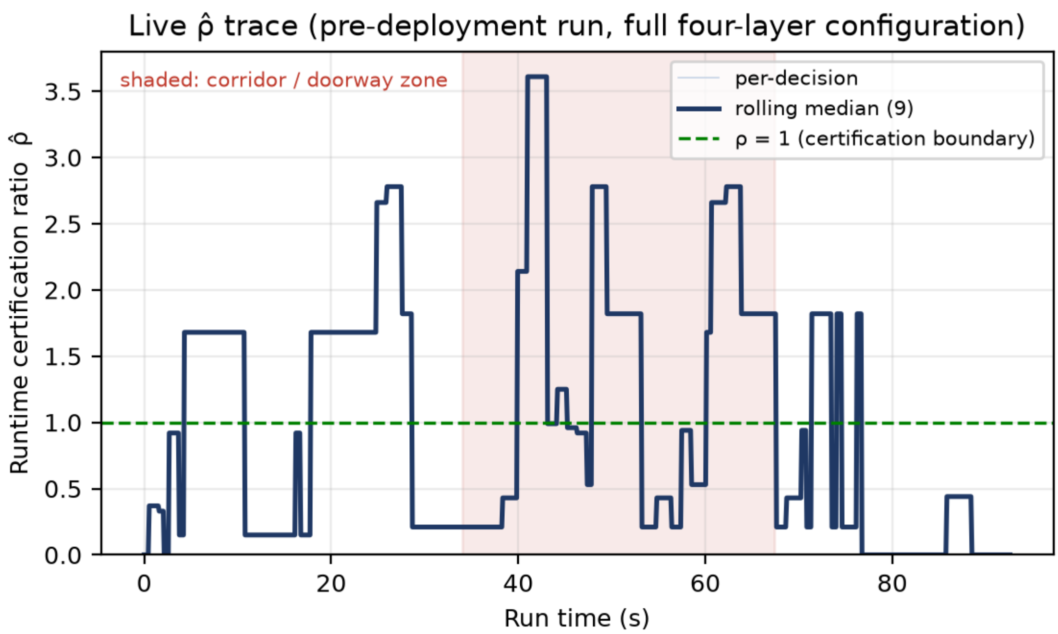

Figure 6. Live $\hat{\rho}$ trace from a pre-deployment run. The ratio is HIGH inside the corridor/doorway (median 1.25: readings are steady, the conservative decision is unambiguous — certified) and LOW in open cruising (median ≈ 0.4: clearance flickers,

DST intervals widen, and the fast-versus-careful arbitration is genuinely fragile — boundary regime). The certificate flags where arbitration is vulnerable, not where the environment is dangerous.

$\hat{\rho}$ certifies **decision robustness**, not situational safety. In the doorway the world is unambiguous and the decision margin dwarfs the perturbation budget; in open stretches the flickering laser widens the few-shot DST margins and the profile arbitration is genuinely contestable — exactly the conservative behaviour the theory prescribes (hysteresis holds the incumbent; FALLBACK bursts concentrate there). The absolute scale of $\hat{\rho}$ depends on the instantiation constants ($B = 1$); calibrating them so that $\hat{\rho} \geq 1$ empirically coincides with decisions that never need reversal is a planned exercise over the physical-robot traces.

6. Consistency with theory: claim-by-claim audit

Every claim the campaign instantiates, with verdict. No claim was found inconsistent; two artefacts were caught by the audit process itself and corrected before affecting conclusions (a run with governance silently disabled, excluded; a single-candidate arm originally mislabelled as the M1 baseline, retained as the HELP-semantics arm).

6.1 Paper (Decentralised Capability Abstraction)

Claim	Evidence	Verdict
Availability ≠ deployability: governance certifies capabilities against current experience, not reachability	Governed arms de-certify the fast capability by context; 0.50 → 0.00 spills/run at +29% time, 8/8 completion (Table 6, Fig. 1)	Confirmed (statistical)
Conservative non-selection: an empty deployable set must yield FALLBACK/HELP, never silent persistence	Single-candidate arm: mobility hard veto at the corridor pinch → sustained HELP → governed abort (3/8); same semantics in 3 further lid-arm runs	Confirmed (semantics observed)
Calibration-limited regime (θ_QoE, Table IV): interface quality bounds decision quality	Physical-lab QoE on the twin leaves the efficient capability non-certifiable; measured distributions → re-calibration restores it (Fig. 3)	Confirmed (observed live)
Robustness certification ρ_DCA separates stable / boundary / non-certifiable regimes (Table V)	Runtime instantiation logged per decision: 1.50 clean vs 0.58 degraded (controlled); live traces certified in unambiguous zones, boundary under flickering readings (Fig. 6)	Confirmed ordinally; absolute scale pending B-calibration on physical traces
Hidden context is handled by compressing it into shared experience, interpreted per capability	Task-blind prior corrected within ≈4 s from per-tick shared-experience readings alone, before any failure materialises	Confirmed (measured)

6.2 Thesis (Chapter 5)

Claim	Evidence	Verdict
Safety ordering M0 < M1 ≈ M2, with M2 adaptive rather than rigid (5.3.6.1, 5.3.6.3.1)	M0: 0.50 spills, 8.1% risk; M1 and M2: 0 spills. M2 alone re-certifies speed when conditions permit (L3 arm: 97.4 s vs M1 110.8 s at equal safety)	Confirmed
Zero-shot task-conditioned prior selection (5.3.6.4 lid flag → covered_delivery)	Sealed-lid governed arm on the calibrated twin: −9 s versus open-cup governance at equal safety; ungoverned baseline gains only passively (Fig. 4)	Confirmed
A wrong prior is recoverable through belief revision, never locked in (5.3.6.2-5.3.6.3)	Layer-2 Dempster fusion: Pl 1.00 → 0.75 → 0.53 → 0.36 under attributed spills; run 4 starts pre-corrected (Fig. 5). In the twin the failure signal is pre-empted by the faster layer; operational regime predicted to be the physical laboratory	Confirmed mechanistically; physical-robot exercise pending
Jerk — not speed alone — dominates spill risk (5.3.1.3)	Slosh model driven by acceleration (arm lever, gait ∝ v^2, impact impulses); commanded smooth stops do not spill, impacts and fast sustained gait do; calibrated ≈2 spills / 80 s at 0.30 m/s sustained	Confirmed by construction + calibration
Four belief layers at separated timescales, each catching a distinct failure mode (5.2.6.3)	All four realised (Table 4); L1/L2 exercised in campaign, L3/L4 validated in-loop (§5.1). The twin measures the L1-dominant regime — the timescale separation itself — with the L2-dominant regime correctly assigned to the physical lab	Confirmed; one regime per platform
Multi-environment generality (domestic/warehouse × clear/cluttered)	One environment (the laboratory twin) with open-versus-corridor contrast only	Not tested in the twin — covered by the 2-D study; physical campaign adds the real lab

7. Why it works: a mechanism-level account

The results are not an emergent accident of tuning; each can be traced to a specific mechanism, and each mechanism to the failure it exists to prevent. Six causal steps carry the whole architecture:

(1) The analogy supplies structure that cannot be learned in time. Spill physics is not observable a priori by any onboard sensor, and exploring it empirically means breaking cups — the thesis's two founding constraints (online policy trapping and open-world scope collapse, §5 intro) hold literally here. The analogy imports a complete evaluative structure from a structurally similar base domain — attentions, QoE expectations, vetoes, a policy ceiling — so the robot starts already knowing that carrying an open liquid means safety dominates urgency. Measured consequence: **zero spills in the first run of all nine governed arms**. No exploration paid for that knowledge.

(2) Shared experience makes hidden context actionable without a world model. Five scalar meta-parameters — clearance, progression, mobility, fragility, reliability — were sufficient for the governance outcome; no fluid model, no map of the cup, no learned dynamics. This is the paper's compression claim made concrete: the hidden variable (slosh state) never appears in the interface, yet its deployability consequence does (through fragility after outcomes, and through the analogy's speed ceiling before them).

(3) DST provides the right epistemology for certification. Readings enter as intervals (belief / plausibility) whose width is earned from measured noise and historical consistency — the few-shot margin. A metric that flickers across a QoE boundary widens its own interval and cannot be certified by a lucky sample; genuine first-run ignorance is represented as uncommitted mass rather than a fabricated probability; and Dempster fusion is order-independent, so run-level evidence accumulates without trajectory artefacts. This is why wrong priors neither lock in (evidence erodes them: Fig. 5) nor flap (consistency is required to certify).

(4) Refusal is a first-class outcome. An optimiser always returns its argmax; a governance layer must be able to return nothing is deployable. The hard-veto path (mobility while pressing, fragility after repeated spills) empties the candidate set and produces sustained HELP and a governed abort — observed 6 times, each a correct refusal, never a silent push. This is the paper's conservative non-selection, and it is what makes the system safe under conditions its designers did not anticipate.

(5) Temporal decomposition assigns each failure mode to the cheapest sufficient corrector. Per-tick evidence corrects a mis-selected prior in ≈ 4 s — before any spill can happen — so the run-level layer never even sees a failure in the twin; the run-level layer catches what per-tick channels are blind to (gait-induced spills at legal clearance, the physical-lab regime); the deployment layer catches profile mismatch that neither can see. The twin campaign measured the first regime and the controlled trace validated the second: the separation of timescales is not an assumption, it is an observation (Table 4, Figs. 5-6).

(6) Every decision is auditable, and the audit works. Each tick logs action, active analogy, per-analogy rejection reasons, tension, fulfilment, plausibilities, environment scale and $\hat{\rho}$. That instrumentation is not cosmetic: it is what detected the one corrupted run (governance silently off — visible as null governance fields), what diagnosed the calibration-limited regime (Fig. 3), and what produced every figure in this document directly from run data. Accountability is a property of the architecture, not of the write-up.

8. Benefits of meta-reasoning with analogy — quantified

- **Safety, robustly: 10.6× fewer spills.** Across ALL 64 governed runs — every prior, both calibrations, both lid states, any layer configuration — 3 spills total (0.047/run) versus 0.50/run ungoverned; risk exposure drops from 8-11% of run time to ≤ 1-2%. The benefit does not depend on choosing the right configuration: it is a property of governance itself.
- **Zero-shot competence: no exploratory cost.** First-run spills across the nine governed arms: 0/9. The analogical prior places the robot in a productive, safe regime from the first second of deployment — the 2-D study's convergence-at-run-1 (5.3.6.3.3), reproduced with physics.
- **Bounded, tunable time cost.** Governance costs +29% time in the plain configuration and +17% with the environment-relevance layer active (97.4 s vs 83.4 s baseline) at equal safety. The overhead is a dial (Layer 3, profile choice), not a tax.
- **Safe to be wrong: prior error is absorbed.** Loading a task-blind prior changed the outcome by +0.3% time and zero spills (108.2 ± 3.5 vs 107.9 ± 4.2). Contrast the unprotected wrong prior in the 2-D study: 12.7× the spill rate and evaluation collapse (SE 37.6 → 1.8). With arbitration plus trust revision, the cost of a wrong analogy is bounded and recoverable; without them it is catastrophic.
- **Principled refusal.** Six governed aborts, each from a sustained empty deployable set; every completed governed run was collision-free. A system that cannot refuse cannot be trusted with a payload.
- **Interpretability end-to-end.** Any decision can be answered with named quantities: which analogy, why the others were rejected (per-analogy reasons), how tense, how fulfilled, how plausible, how certified ($\hat{\rho}$). The corrupted-run detection and the calibration diagnosis were performed from logs alone, in minutes.
- **Configuration-first transfer.** Lab → twin adaptation was four JSON numbers derived from measured percentiles (no code, no retraining); task transfer (open → sealed cup) is selecting a configuration; the reasoner was never modified across 103 runs, two platforms and twelve arms.
- **Runtime economy.** Pure-Python reasoner at 2 Hz, no GPU, no training loop, unchanged from consumer-robot hardware to simulator — deployable on anything that can run the stack.

9. Limitations and threats to validity

- **Kinematic twin.** The robot is a velocity-controlled body with synthesized gait excitation, not simulated biped dynamics; contacts appear as pressing/stalls rather than impacts at speed, so the impact-impulse spill channel is under-exercised in simulation (it will be exercised physically).
- **Spill model plausibility, not fluid-dynamics validation.** Parameters are physically motivated and calibrated for plausible rates; the manual ground-truth channel on the real robot provides the external check.
- **One environment, n = 8 per arm.** Zero-spill contrasts are unambiguous; time contrasts within ± 5 s should not be over-read. Multi-environment generality rests on the 2-D study; the physical campaign adds the real laboratory.
- **Real-time factor ≈ 0.5.** Wall-clock durations are dilated ~2× relative to sim time, identically across arms; escalation windows were scaled accordingly and documented.
- **Camera channel disabled in the twin.** Governance inputs are LiDAR-derived; the physical robot adds the vision channels (floor-colour, depth, detection) to the same interface.

- **$\hat{\rho}$ absolute scale uncalibrated.** Ordinal behaviour matches theory; choosing B constants so that $\hat{\rho} \geq 1$ coincides empirically with never-reversed decisions is planned over physical traces.
- **Layers 3-4 in-loop evidence is 4 runs.** Sufficient for no-regression and correct direction of effect; their statistics are deliberately excluded from the published arm table.

Datasets, configurations, the aggregation script and figure-generation code are versioned with the robot stack (commits through the 2026-07-04 campaigns; rev. 4).